Bent Marshak Waves: Axisymmetric Cylindrical Model

Eigenvalues, Longitudinal Calibration, and Front Curvature

Yuval Kochavi

Updated October 2026

1 Physical setup and assumptions

We consider an optically thick, stationary foam within a cylindrical sleeve of radius R. The axial direction is z, the radial direction is $r \in [0, R]$, and axial symmetry implies $\partial_\theta = 0$. A radiation-temperature drive $T_s(t)$ is imposed at $z = 0$. We take constant foam density ρ and constant Rosseland opacity κ; the wall has albedo a. Let

$$U(z, r, t) = T^4(z, r, t), \qquad U_s(t) = T_s^4(t). \tag{1.1}$$

As in the quasi-steady region of Hurricane and Hammer's *Bent Marshak Waves* (2006), the temperature field is treated as harmonic behind a steep front. This is not a full time-dependent diffusion solution, and corrections to the front position should be distinguished from a self-consistent moving-boundary solution.

2 Governing equation and cylindrical modes

The diffusion approximation gives

$$\rho \frac{\partial e}{\partial t} = \frac{4}{3} \nabla \cdot \left[\frac{\sigma}{\rho \kappa} \nabla U \right], \tag{2.1}$$

where e is *specific* internal energy (energy per mass) and σ is the Stefan–Boltzmann constant in the paper's units. Assuming $\partial_t e$ is negligible away from the front, and constant ρ, κ,

$$\boxed{\partial_z^2 U + \frac{1}{r} \partial_r (r \partial_r U) = 0.} \tag{2.2}$$

With $U = Z(z)F(r)$, separation yields

$$Z'' - k^2 Z = 0, \qquad F'' + \frac{1}{r} F' + k^2 F = 0. \tag{2.3}$$

The finite radial solution is $F(r) = J_0(kr)$; $Y_0(kr)$ is singular on the axis. The full modal solution is

$$\boxed{U(z, r, t) = \sum_{n=0}^{\infty} J_0(k_n r) \left[A_n(t) e^{k_n z} + B_n(t) e^{-k_n z} \right].} \tag{2.4}$$

3 Wall boundary condition and eigenvalues

The axis is regular, $\partial_r U(0,z,t)=0$. The outward radiative loss at $r=R$ gives the Robin condition

$$\partial_r U|_R = -hU(R,z,t), \qquad h=\frac{3}{4}\rho\kappa(1-a). \tag{3.1}$$

Introduce $\varepsilon = hR = \frac{3}{4}\rho\kappa R(1-a)$ and $\lambda_n = k_n R$. Since $J_0'(x) = -J_1(x)$, each mode obeys

$$\boxed{\lambda_n J_1(\lambda_n) = \varepsilon J_0(\lambda_n), \qquad k_n = \lambda_n/R.} \tag{3.2}$$

This replaces the planar relation $\tan(k_n L) = \varepsilon/(k_n L)$. For $\varepsilon > 0$, the roots $\lambda_n > 0$ are ordered increasingly; $\lambda_0 \to 0$ as $\varepsilon \to 0$.

3.1 Small-wall-loss eigenvalue expansions

The Bessel series are

$$J_0(x) = 1 - \frac{x^2}{4} + \frac{x^4}{64} + O(x^6), \tag{3.3}$$

$$J_1(x) = \frac{x}{2} - \frac{x^3}{16} + \frac{x^5}{384} + O(x^7). \tag{3.4}$$

For the fundamental mode,

$$\varepsilon = \frac{\lambda_0 J_1(\lambda_0)}{J_0(\lambda_0)} = \frac{\lambda_0^2}{2} + \frac{\lambda_0^4}{16} + \frac{\lambda_0^6}{96} + O(\lambda_0^8), \tag{3.5}$$

so

$$\boxed{\lambda_0 = \sqrt{2\varepsilon}\left(1 - \frac{\varepsilon}{8} + \frac{5\varepsilon^2}{384} + O(\varepsilon^3)\right).} \tag{3.6}$$

For $n \geq 1$, let $\alpha_n = j_{1,n}$ be the n-th positive zero of J_1 and write $\lambda_n = \alpha_n + \delta_n$. Expanding (3.2) around α_n and using $J_1'(\alpha_n) = J_0(\alpha_n)$ gives

$$\boxed{\lambda_n = \alpha_n + \frac{\varepsilon}{\alpha_n} - \frac{\varepsilon^2}{2\alpha_n^3} + O(\varepsilon^3), \quad n \geq 1.} \tag{3.7}$$

The first positive zeros are $\alpha_1 \simeq 3.831706$, $\alpha_2 \simeq 7.015587$, $\alpha_3 \simeq 10.173468$.

3.2 Weighted orthogonality and exact source coefficients

Define $\phi_n(r) = J_0(k_n r)$. The eigenfunctions are orthogonal with cylindrical weight r:

$$\int_0^R r\phi_n(r)\phi_m(r)\,dr = N_n\delta_{nm}, \qquad N_n = \frac{R^2}{2}\left[J_0^2(\lambda_n) + J_1^2(\lambda_n)\right]. \tag{3.8}$$

The simpler norm using only J_1 applies to Dirichlet zeros of J_0 and must *not* be used for the present Robin eigenvalues. For a radially uniform boundary $U(0,r,t) = U_s(t)$, let $1 = \sum_{n\geq 0} b_n\phi_n(r)$. Then

$$\boxed{b_n = \frac{2J_1(\lambda_n)}{\lambda_n[J_0^2(\lambda_n) + J_1^2(\lambda_n)]}.} \tag{3.9}$$

For $\varepsilon \ll 1$, $b_0 = 1 + \varepsilon/4 + O(\varepsilon^2)$, while $b_{n\geq 1} = O(\varepsilon)$. The source cannot be represented *exactly* by the fundamental mode alone for nonzero wall loss.

4 Longitudinal conditions and quasi-steady temperature profile

The source condition is $U(0,r,t) = U_s(t)$. First solve an auxiliary *flat-front* boundary problem with a reference plane $z = \ell(t)$ and $U(\ell,r,t) = 0$. Modal separation and (3.9) then give

$$\boxed{\frac{U(z,r,t)}{U_s(t)} = \sum_{n=0}^{\infty} b_n J_0(k_n r) \frac{\sinh[k_n(\ell(t)-z)]}{\sinh[k_n \ell(t)]}.} \tag{4.1}$$

This field meets both longitudinal boundary conditions (up to the usual convergence interpretation at the source–wall corner). The leading-mode approximation is

$$\frac{U}{U_s} \simeq b_0 J_0(k_0 r) \frac{\sinh[k_0(\ell - z)]}{\sinh(k_0 \ell)}. \tag{4.2}$$

The planar-front approximation is an intermediate computational closure: it should not be mistaken for an exact solution satisfying $U(z_F(r,t),r,t) = 0$ at a curved front. Terms from the front displacement enter upon Taylor-expanding $U(z_F,r,t)$ about $z = \ell$.

5 Illustrative lowest-mode front evolution

For a steep front, the approximate axial energy balance is

$$\rho e \, \partial_t z_F = -\frac{4\sigma}{3\rho\kappa} \, \partial_z U|_{\text{front}} \,, \tag{5.1}$$

where e is the specific energy required to heat initially cold foam. Projecting the lowest-mode gradient of (4.2) onto the lowest mode, and taking the *model* front $z_F^{\text{mod}}(r,t) = c_0(t) J_0(k_0 r)$ with the auxiliary plane $\ell = c_0$, yields

$$\dot{c}_0 = \frac{D_M(t) b_0 k_0}{2 \sinh(k_0 c_0)}, \qquad D_M(t) = \frac{8\sigma U_s(t)}{3 e \rho^2 \kappa}. \tag{5.2}$$

This is a leading-mode approximation; it is not an exact pointwise moving-front solution. With $c_0(0) = 0$, its integral is

$$\boxed{c_0(t) = \frac{1}{k_0} \operatorname{arcosh}\left[1 + \frac{b_0 k_0^2}{2} \int_0^t D_M(s)\, ds\right].} \tag{5.3}$$

For a constant drive, D_M is constant and the integral equals $D_M t$. At leading order $b_0 \simeq 1$; retaining b_0 gives the source-normalization correction, $b_0 = 1 + \varepsilon/4 + O(\varepsilon^2)$, not an independently established $1 + \varepsilon/2$ correction. Higher-order consistency requires retaining higher radial modes and curved-front terms (Section 7).

6 Calibration to the correct on-axis longitudinal position

The front may be predicted inaccurately along z even if the transverse *bending* is useful. To avoid confusing the correct on-axis position with the modal estimate, define

$$Z_F(t) = \text{correct or externally supplied on-axis front position}, \qquad z_F^{\text{mod}}(r,t) = \sum_{n=0}^{\infty} c_n(t) J_0(k_n r). \tag{6.1}$$

As $J_0(0) = 1$, $z_F^{\rm mod}(0,t) = \sum_n c_n(t)$. A *pure longitudinal correction*, preserving the original transverse displacement relative to the axis, is

$$\boxed{\widetilde{z}_F(r,t) = Z_F(t) + \sum_{n=0}^{\infty} c_n(t)\,[J_0(k_n r) - 1]\,.} \tag{6.2}$$

Indeed $\widetilde{z}_F(0,t) = Z_F(t)$. This is equivalent to translating the model front by $Z_F - \sum_n c_n$. It is a geometric calibration, not a new solution of (5.1) or of the complete nonlinear boundary-value problem. It also does not, by itself, correct the underlying temperature field.

6.1 On-axis curvature and curvature radius

For the meridional curve $z = \widetilde{z}_F(r,t)$, the unsigned curvature is

$$\mathcal{K}_m(r,t) = \frac{|\partial_r^2 \widetilde{z}_F|}{[1 + (\partial_r \widetilde{z}_F)^2]^{3/2}}. \tag{6.3}$$

By axisymmetry, $\partial_r \widetilde{z}_F(0,t) = 0$. Using

$$J_0(k_n r) = 1 - \frac{k_n^2 r^2}{4} + O(r^4), \tag{6.4}$$

one obtains (assuming sufficient convergence to differentiate the modal series)

$$\widetilde{z}_F(r,t) = Z_F(t) - \frac{r^2}{4R^2} \sum_{n\geq 0} c_n(t)\lambda_n^2 + O(r^4), \tag{6.5}$$

$$\partial_r^2 \widetilde{z}_F\Big|_{r=0} = -\frac{1}{2R^2} \sum_{n\geq 0} c_n(t)\lambda_n^2. \tag{6.6}$$

Therefore the *on-axis meridional curvature radius* is

$$\boxed{\mathcal{R}_c(t) = \frac{2R^2}{|\sum_{n=0}^{\infty} c_n(t)\lambda_n^2|}.} \tag{6.7}$$

At the axis, the two principal curvatures of the smooth surface of revolution agree in magnitude; away from the axis the meridional and azimuthal principal curvatures generally differ. Most importantly, (6.7) is *independent* of $Z_F(t)$ for an additive longitudinal correction.

6.2 Fundamental mode and alternative rescaling

Keeping only $n = 0$ in (6.2),

$$\widetilde{z}_F(r,t) \simeq Z_F(t) + c_0(t)[J_0(k_0 r) - 1], \qquad \boxed{\mathcal{R}_c(t) \simeq \frac{2R^2}{|c_0(t)|\lambda_0^2} \simeq \frac{R^2}{\varepsilon |c_0(t)|}.} \tag{6.8}$$

If instead one decides to rescale the *entire fundamental-mode amplitude* to match Z_F, the different assumption is

$$\widetilde{z}_F^{\rm scale}(r,t) = Z_F(t) J_0(k_0 r), \qquad \boxed{\mathcal{R}_c^{\rm scale}(t) = \frac{2R^2}{|Z_F(t)|\lambda_0^2} \simeq \frac{R^2}{\varepsilon |Z_F(t)|}.} \tag{6.9}$$

The two formulas are *not interchangeable*: translation preserves curvature; amplitude rescaling changes it. A more general amplitude adjustment $A(t)$ multiplying the bracketed sum in (6.2) scales the on-axis curvature by $|A(t)|$.

7 Higher-order curvature when the on-axis position is prescribed

Throughout this section the on-axis trajectory $Z_F(t)$ is regarded as *known* (from a 1D Marshak model, a simulation, or a measurement). Only the transverse shape remains to be determined:

$$z_F(r,t) = Z_F(t) + \eta(r,t), \qquad \eta(0,t) = 0, \qquad \mathcal{K}(t) \equiv \frac{1}{\mathcal{R}_c(t)} = |\partial_r^2 \eta(0,t)|. \tag{7.1}$$

The aim is to obtain $\mathcal{K}$ beyond the single-mode result (6.8) without re-predicting Z_F.

7.1 Closure: Z_F fixes the magnitude, the model fixes the ratio

For a curved front, $U = 0$ along $z = z_F(r,t)$, so ∇U is normal to the front and its tangential derivative vanishes, $\partial_r U + \partial_r z_F \, \partial_z U = 0$. Hence $|\nabla U| = q\sqrt{1 + (\partial_r z_F)^2}$ with

$$q(r,t) \equiv -\left.\partial_z U\right|_{z=z_F(r,t)}. \tag{7.2}$$

The normal front speed is $\partial_t z_F / \sqrt{1 + (\partial_r z_F)^2}$, so the curved-front generalization of (5.1) is

$$\rho e \, \partial_t z_F = \frac{4\sigma}{3\rho\kappa}\left[1 + (\partial_r z_F)^2\right] q(r,t). \tag{7.3}$$

At $r = 0$ the slope vanishes and $\partial_t z_F = \dot{Z}_F$. Dividing (7.3) by its on-axis value removes ρe, σ, κ, D_M, the drive, and the source normalization b_0:

$$\boxed{\partial_t z_F(r,t) = \dot{Z}_F(t)\left[1 + (\partial_r z_F)^2\right] \frac{q(r,t)}{q(0,t)}.} \tag{7.4}$$

This is the natural way to use a prescribed Z_F: its *rate* supplies the magnitude of the motion, and the model only supplies the *radial ratio* of front speeds. Using Z_F itself as the clock,

$$\boxed{\frac{\partial \eta}{\partial Z_F} = \left[1 + (\partial_r \eta)^2\right] \frac{q(r)}{q(0)} - 1, \qquad \left.\eta\right|_{Z_F = 0} = 0.} \tag{7.5}$$

Two consequences follow. First, any error in the absolute axial speed of the model is irrelevant. Second, for constant ε the shape depends on Z_F/R and ε only, not on the drive history.

Leading order. With the single-mode, flat-plane field (4.2) evaluated at $\ell = Z_F$, $q(r)/q(0) = J_0(k_0 r)$. Dropping the $O(\varepsilon^2)$ slope factor, (7.5) integrates to

$$z_F(r,t) = Z_F(t)\, J_0(k_0 r), \qquad \mathcal{K}^{(0)} = \frac{\lambda_0^2 Z_F}{2R^2} \simeq \frac{\varepsilon Z_F}{R^2}. \tag{7.6}$$

This is exactly the *rescaled* variant of Section 6, and it is the profile currently implemented in the code (`_compute_z_front_radial_snapshot` in `model_main.py`, $z_F(r) = x_F\, J_0(k_0 r)$). The closure (7.4) therefore selects rescaling, not translation, when Z_F is trusted. Every higher-order term below is a correction to $q(r)/q(0)$ in (7.5).

7.2 Bookkeeping: which parameter is small

The only small parameter is the wall-loss number ε; the penetration ratio Z_F/R is $O(1)$ and must *not* be expanded. All results take the form

$$\mathcal{K} = \frac{\varepsilon}{R}\,\mathcal{F}_1\!\left(\frac{Z_F}{R}\right) + \frac{\varepsilon^2}{R}\,\mathcal{F}_2\!\left(\frac{Z_F}{R}\right) + O(\varepsilon^3). \tag{7.7}$$

The single-mode model gives $\mathcal{F}_1 = Z_F/R$. Corrections 2 and 3 below show that the higher radial modes and the curved-front boundary condition both change $\mathcal{F}_1$ at *leading* order; they are not small refinements. Only Correction 1 and the slope factor in (7.5) are genuinely $O(\varepsilon^2)$.

At $O(\varepsilon)$ the Robin modes may be replaced by Neumann modes. Writing $\alpha_n = j_{1,n}$ and $\psi_n(r) = J_0(\alpha_n r/R)$, the source coefficients of Section 3 reduce to

$$b_n \simeq \frac{2\varepsilon}{\alpha_n^2 J_0(\alpha_n)} \quad (n \geq 1). \tag{7.8}$$

The expansion of r^2, which vanishes at the axis, is

$$r^2 = \sum_{n\geq 1} \frac{4R^2}{\alpha_n^2 J_0(\alpha_n)}\big[\psi_n(r) - 1\big], \qquad \text{so} \qquad J_0(k_0 r) - 1 = -\sum_{n\geq 1} b_n\big[\psi_n(r) - 1\big] + O(\varepsilon^2). \tag{7.9}$$

Any η with $\eta(0) = 0$ is expanded as $\eta = \sum_{n\geq 1} \eta_n(\psi_n - 1)$, which gives

$$\partial_r^2 \eta(0) = -\sum_{n\geq 1} \frac{\alpha_n^2}{2R^2}\eta_n. \tag{7.10}$$

The series below are alternating and converge slowly. The constant

$$C_\infty \equiv \sum_{n\geq 1} \frac{1}{\alpha_n J_0(\alpha_n)} = -0.38479\ldots \tag{7.11}$$

recurs throughout. Numerically, subtract it and sum only the exponentially convergent remainders.

7.3 Correction 1: the eigenvalue at higher order in ε

Inverting the series of Section 3.1,

$$\lambda_0^2 = 2\varepsilon\left(1 - \frac{\varepsilon}{4} + \frac{\varepsilon^2}{24} + O(\varepsilon^3)\right) \;\Rightarrow\; \mathcal{K}^{(0)} = \frac{\varepsilon Z_F}{R^2}\left(1 - \frac{\varepsilon}{4} + \frac{\varepsilon^2}{24}\right). \tag{7.12}$$

This is a pure ε^2 correction (about 7% at $\varepsilon = 0.3$). It is included automatically whenever λ_0 is the exact root of (3.2), as in `kappa_roots`. It does *not* change the Z_F dependence.

7.4 Correction 2: higher radial modes on the flat auxiliary plane

Evaluating the full series (4.1) at $\ell = Z_F$ gives

$$\frac{q(r)}{q(0)} = \frac{\sum_n w_n J_0(k_n r)}{\sum_n w_n}, \qquad w_n(Z) = \frac{b_n \lambda_n}{\sinh(\lambda_n Z/R)}. \tag{7.13}$$

Inserting this into (7.5) gives, within this closure and for any ε,

$$\mathcal{K}_{\text{plane}} = \frac{1}{2R^2}\int_0^{Z_F} \langle\lambda^2\rangle(Z)\,dZ, \qquad \langle\lambda^2\rangle = \frac{\sum_n w_n\lambda_n^2}{\sum_n w_n}. \tag{7.14}$$

At $O(\varepsilon)$, use (7.8)–(7.9) and $x_n \equiv \alpha_n Z_F/R$:

$$\boxed{\mathcal{K}_{\text{plane}} = \frac{\varepsilon}{R^2}\left|Z_F + R\sum_{n\ge1}\frac{I(x_n)}{\alpha_n J_0(\alpha_n)}\right|, \qquad I(x) = \int_0^x \frac{u\,du}{\sinh u} \xrightarrow{x\to\infty} \frac{\pi^2}{4}.} \tag{7.15}$$

For $Z_F \gtrsim 2R$ the remainders are exponentially small, giving an *effective origin shift*

$$\mathcal{K}_{\text{plane}} \simeq \frac{\varepsilon}{R^2}(Z_F - Z_*), \qquad Z_* = -\frac{\pi^2 R}{4}C_\infty \simeq 0.949\,R. \tag{7.16}$$

At finite ε, using $\int_0^\infty \sinh(az)/\sinh(bz)\,dz = (\pi/2b)\tan(\pi a/2b)$,

$$Z_* = -\frac{\pi R}{2b_0\lambda_0^3}\sum_{n\ge1} b_n(\lambda_n^2 - \lambda_0^2)\tan\frac{\pi\lambda_0}{2\lambda_n}. \tag{7.17}$$

This gives $Z_*/R = 0.943$, 0.937, and 0.916 for $\varepsilon = 0.05$, 0.1, and 0.3.

Physically, near the source the wall's influence has not yet reached the axis, so the front starts flat; the higher modes represent this delay. The $O(\varepsilon)$ coefficients $b_{n\ge1}$ are small, but they are multiplied by λ_n^2 and by an integration over Z, so the shift is $O(R)$, not $O(\varepsilon R)$. For $Z_F \lesssim R$ this closure is dominated by the source–wall corner, where the quasi-steady model is least reliable.

7.5 Correction 3: the curved-front boundary condition

Both previous results evaluate q on the plane $z = Z_F$, whereas U must vanish on $z = Z_F + \eta(r)$. Write $U = U^{(0)} + U^{(1)}$, where $U^{(0)}$ is the plane solution (4.1) with $\ell = Z_F$, and let $q^{(0)} = -\partial_z U^{(0)}(Z_F, r)$. Taylor-transferring the condition to the plane gives

$$U^{(1)}(Z_F, r) = \eta(r)\,q^{(0)}(r), \qquad U^{(1)}(0,r) = 0, \qquad U^{(1)} = \sum_n g_n J_0(k_n r)\frac{\sinh k_n z}{\sinh k_n Z_F}, \tag{7.18}$$

with $g_n = N_n^{-1}\int_0^R r\,\eta\,q^{(0)}J_0(k_n r)\,dr$. Because $U^{(0)} \equiv 0$ on the plane, $\partial_z^2 U^{(0)} = -r^{-1}\partial_r(r\partial_r U^{(0)}) = 0$ there. The flux on the true front is therefore

$$\boxed{q(r) = q^{(0)}(r) - \sum_n g_n\,k_n\coth(k_n Z_F)\,J_0(k_n r) + O(\eta^2).} \tag{7.19}$$

The second term is a Dirichlet-to-Neumann response. A part of the front that lags by η_n in radial mode n is closer to the source, so it receives an extra flux of relative size $\approx k_n|\eta_n|$ and catches up: the front is *self-flattening*.

This term is not optional. In the plane closure $\eta \sim \varepsilon Z_F$, so the relaxation $k_1\eta \sim \varepsilon\alpha_1 Z_F/R$ is of the same order as the wall-loss forcing ($\sim\varepsilon$) as soon as $Z_F \gtrsim R/\alpha_1 \approx 0.26R$.

Leading-order solution. At $O(\varepsilon)$, $q^{(0)} \simeq U_s/Z_F$ inside g_n, so $g_n \simeq (U_s/Z_F)\eta_n$; the $n=0$ term only contributes at $O(\varepsilon^2)$. Expanding $q^{(0)}(r)/q^{(0)}(0) - 1$ with (7.9) gives the forcing $\sigma_n = -b_n\,(1 - x/\sinh x)|_{x=\alpha_n Z/R}$. Equation (7.5) then decouples into independent linear relaxation equations,

$$\frac{d\eta_n}{dZ_F} = -b_n\left(1 - \frac{x_n}{\sinh x_n}\right) - \frac{\alpha_n}{R}\coth(x_n)\,\eta_n, \qquad x_n = \frac{\alpha_n Z_F}{R}. \tag{7.20}$$

With the integrating factor $\sinh x_n$ they have the closed-form solution

$$\eta_n = -\frac{b_n R}{\alpha_n}\,F(x_n), \qquad F(x) = \frac{\cosh x - 1 - x^2/2}{\sinh x} \xrightarrow{x\to\infty} 1, \qquad F(x) \simeq \frac{x^3}{24}\ (x \ll 1). \tag{7.21}$$

Hence the front shape and the on-axis curvature are

$$z_F(r,t) = Z_F(t) - 2\varepsilon R\sum_{n\ge1}\frac{F(x_n)}{\alpha_n^3 J_0(\alpha_n)}\Big[J_0\big(\tfrac{\alpha_n r}{R}\big) - 1\Big], \tag{7.22}$$

$$\mathcal{K}_{\rm sc} = \frac{\varepsilon}{R}\left|\sum_{n\ge1}\frac{F(x_n)}{\alpha_n J_0(\alpha_n)}\right| = \frac{\varepsilon}{R}\left|C_\infty + \sum_{n\ge1}\frac{F(x_n)-1}{\alpha_n J_0(\alpha_n)}\right|. \tag{7.23}$$

For $Z_F \gtrsim 2R$ the curvature *saturates*:

$$\boxed{\mathcal{K}_\infty = \frac{|C_\infty|\,\varepsilon}{R} \simeq 0.385\,\frac{\varepsilon}{R}, \qquad \mathcal{R}_{c,\infty} \simeq 2.60\,\frac{R}{\varepsilon}, \qquad Z_F - z_F(R) \to 0.120\,\varepsilon R.} \tag{7.24}$$

This is a balance between wall-loss forcing and lateral relaxation, independent of Z_F. It differs qualitatively from (7.6), where $\mathcal{R}_c \propto 1/Z_F$.

The solution also satisfies the corner condition automatically. Since $J_1(\alpha_n) = 0$, $\partial_r\eta(R) = 0$, which is required at a regular point where $U = 0$ meets a Robin wall: there $\partial_r U = -hU = 0$, so the tangential condition gives $\partial_r z_F\,\partial_z U = 0$. The single-mode profile (7.6) instead has wall slope $-Z_F k_0 J_1(\lambda_0) \simeq -\varepsilon Z_F/R$.

Time-dependent wall loss. Because the equations are linear and $b_n \propto \varepsilon$, a slowly varying $\varepsilon(t)$ (e.g. from a time-dependent albedo) is included by superposition. Let $\varepsilon(Z')$ denote its value when the axis front was at Z'. Then

$$\partial_r^2\eta(0,t) = \frac{1}{R^2}\sum_{n\ge1}\frac{1}{J_0(\alpha_n)\sinh x_n(t)}\int_0^{Z_F(t)}\varepsilon(Z')\left[\sinh\frac{\alpha_n Z'}{R} - \frac{\alpha_n Z'}{R}\right]dZ'. \tag{7.25}$$

The kernel decays as $e^{-\alpha_n(Z_F - Z')/R}$, so the memory length is $R/\alpha_1 \approx 0.26R$. Using the instantaneous $\varepsilon(t)$ is therefore justified here. In the plane closure, by contrast, the leading term becomes $R^{-2}\int_0^{Z_F}\varepsilon\,dZ'$ and the full history matters.

7.6 Comparison and numerical check

To test the expansions, (7.5) was also solved without linearization. The field is represented as $U = U_s\sum_{n<N} J_0(k_n r)\big[b_n\sinh k_n(Z_F - z) + \beta_n\sinh k_n z\big]/\sinh k_n Z_F$, which satisfies the source and Robin conditions exactly. The β_n are found by weighted least squares so that $U = 0$ on the current curved front, and the front is advanced in Z_F (RK2, $N = 120$ exact Robin modes, start flat at $Z_F = 0.02R$). The residual on the front stays below $10^{-3}U_s$, and the results change by about 3% when N is lowered to 70. This reference includes all orders in ε and η except the slope factor, which is $O(10^{-2}\varepsilon^2)$ there.

	$O(\varepsilon)$ theory: $\mathcal{K}R/\varepsilon$			full curved-front solution: $\mathcal{K}R/\varepsilon$		
Z_F/R	single (7.6)	plane (7.15)	curved (7.23)	$\varepsilon = 0.05$	$\varepsilon = 0.1$	$\varepsilon = 0.3$
0.5	0.50	0.006	0.012	0.011	0.011	0.011
1	1.00	0.180	0.171	0.168	0.165	0.155
2	2.00	1.056	0.366	0.358	0.353	0.331
3	3.00	2.051	0.384	0.374	0.368	0.347
4	4.00	3.051	0.385	0.374	0.368	0.345

The plane column was also reproduced by the same numerical integration with q taken on the plane, and the predicted wall lag $0.120\,\varepsilon R$ matches the numerical value. The remaining difference between the curved-front columns is the $O(\varepsilon^2)$ term $\mathcal{F}_2$, roughly $-(0.3\text{–}0.4)\varepsilon$ in relative terms. Substituting exact Robin λ_n, b_n into (7.23) recovers only about a third of it.

7.7 Off-axis curvature

For (7.22), with $\rho_n = \alpha_n r/R$ and $J_0' = -J_1$, $J_1'(x) = J_0(x) - J_1(x)/x$,

$$\partial_r\eta = 2\varepsilon \sum_{n\geq 1} \frac{F(x_n)J_1(\rho_n)}{\alpha_n^2 J_0(\alpha_n)}, \qquad \partial_r^2\eta = \frac{2\varepsilon}{R} \sum_{n\geq 1} \frac{F(x_n)\big[J_0(\rho_n) - J_1(\rho_n)/\rho_n\big]}{\alpha_n J_0(\alpha_n)}. \tag{7.26}$$

For the single-mode profile, $\partial_r\eta = -Z_F k_0 J_1(k_0 r)$ and $\partial_r^2\eta = -Z_F k_0^2\big[J_0(k_0 r) - J_1(k_0 r)/(k_0 r)\big]$. The two principal curvatures of the surface of revolution are

$$\mathcal{K}_m(r) = \frac{|\partial_r^2\eta|}{[1 + (\partial_r\eta)^2]^{3/2}}, \qquad \mathcal{K}_\theta(r) = \frac{|\partial_r\eta|}{r\,[1 + (\partial_r\eta)^2]^{1/2}}, \tag{7.27}$$

which coincide at $r = 0$. The slope corrections in the denominators are $O(\varepsilon^2)$.

7.8 Next order, $O(\varepsilon^2)$: what must be added

A systematic $\mathcal{F}_2$ requires all of the following together; keeping only some of them is not consistent:

1. exact Robin λ_n, b_n, $J_0(k_n r)$, and N_n in place of α_n, (7.8), and ψ_n, including the slow $n = 0$ relaxation rate $k_0 \coth(k_0 Z_F)$;
2. the radial variation of $q^{(0)}$ inside g_n, i.e. the product $\eta\, q^{(0)}$, not $\eta\, q^{(0)}(0)$;
3. the second-order transfer of the front condition, $U^{(2)}(Z_F, r) = -\eta\, \partial_z U^{(1)}(Z_F, r)$ (the $\frac{1}{2}\eta^2\partial_z^2 U^{(0)}$ term vanishes);
4. the slope factor $1 + (\partial_r\eta)^2$ in (7.5);
5. if $\varepsilon = \varepsilon(t)$, the time dependence of the eigenbasis itself.

The collocation solver above contains items 1–3 automatically and is the practical route beyond $O(\varepsilon)$.

7.9 Recipe for a given $Z_F(t)$

1. Take $Z_F(t)$ from the trusted source and $\varepsilon(t)$ from the albedo model, $\varepsilon = \frac{3}{4}\rho\kappa R(1-a)$.
2. Form $x_n = \alpha_n Z_F/R$ for $n = 1, \ldots, N$. Because $F(x_n) - 1$ and $I(x_n) - \pi^2/4$ decay like $x_n e^{-x_n}$, a few modes suffice once the constant C_∞ is used for the tail.
3. Choose the closure. Plane (Hurricane–Hammer-type): (7.15). Self-consistent curved front: (7.23). Then $\mathcal{R}_c = 1/\mathcal{K}$, with the shape from (7.22) if needed.
4. For $\varepsilon \gtrsim 0.2$, or to quantify $\mathcal{F}_2$, solve (7.5) numerically with the curved-front collocation.

7.10 Implementation in the code

The closures are implemented in `front_curvature.py` (class `FrontShapeModel`) and selected by `Front_curvature_closure` in `parameters.py` (`"single"`, `"plane"`, `"curved"`). The model is advanced once per time step of the Marshak march, with ε and k_0 frozen over each step ΔZ_F. Three choices go beyond the $O(\varepsilon)$ formulas above.

1. *Exact forcing projection.* The forcing $J_0(k_0 r) - 1 = -\sum_{n\geq 1} \tilde{b}_n[\psi_n(r) - 1]$ is projected exactly onto the fixed Neumann basis:

$$\tilde{b}_n = \frac{2\lambda_0 J_1(\lambda_0)}{(\alpha_n^2 - \lambda_0^2)\, J_0(\alpha_n)} = \frac{2\varepsilon J_0(\lambda_0)}{(\alpha_n^2 - \lambda_0^2)\, J_0(\alpha_n)}. \tag{7.28}$$

   This reduces to (7.8) as $\varepsilon \to 0$ and stays bounded for $\varepsilon \gg 1$, as happens at early times when the front is still thin. With it, the curved closure gives $\mathcal{K}R/\varepsilon = 0.379$, 0.373, and 0.352 at $Z_F = 4R$ for $\varepsilon = 0.05$, 0.1, and 0.3. The full curved-front solution gives 0.374, 0.368, and 0.345, so most of the $O(\varepsilon^2)$ gap is removed.

2. *History in the plane closure.* For time-dependent ε the leading term is $\int_0^{Z_F}[J_0(k_0(Z')r) - 1]\, dZ'$, not $Z_F[J_0(k_0 r) - 1]$. It is accumulated through the moments $S_m = \int_0^{Z_F} (k_0 R)^{2m} dZ'$ of the J_0 power series. The mode corrections obey $d\delta_n/dZ_F = \tilde{b}_n\, x_n/\sinh x_n$.

3. *Exact step integration in the curved closure.* The step uses $d(\eta_n \sinh x_n)/dZ_F = -\tilde{b}_n(\sinh x_n - x_n)$, written in overflow-safe form. For constant ε this reproduces (7.22) independently of the step size.

In both closures the truncated series for $\partial_r^2 \eta(0)$ is completed analytically with the remainder of C_∞, weighted by $\lambda_0 J_1(\lambda_0)$. With `Front_shape_feedback`, the wall position $z_F(R)$ of the selected closure replaces $Z_F J_0(\lambda_0)$ in the albedo, heating-time, and wall-penetration models.

7.11 Physical caveat

The self-flattening in Correction 3 is a property of the quasi-steady harmonic model, in which conduction just behind the front relaxes lateral perturbations efficiently. A temperature-dependent opacity alone does not remove it: in the quasi-steady limit it can be absorbed into a Kirchhoff potential that is again harmonic. However, it makes the Robin condition nonlinear, with relatively stronger loss near the cold front. More importantly, the neglected heat-capacity term $\rho\, \partial_t e$ is not small in a supersonic Marshak wave.

Hence (7.6)–(7.15) (curvature keeps growing with Z_F) and (7.23) (curvature saturates at $\mathcal{R}_c \simeq 2.6R/\varepsilon$) should be regarded as two limiting closures of the same leading-order physics, not as a

converged prediction. The observed behaviour of $\mathcal{R}_c$ versus Z_F at fixed ε discriminates between them directly. A 2D simulation or measurement that keeps bending as $\mathcal{R}_c \propto 1/Z_F$ indicates that lateral relaxation is weaker than the harmonic model assumes.

8 Physical interpretation and limitations

The cylindrical eigenfunctions and Robin eigenvalues determine the radial dependence associated with wall losses. The leading eigenvalue $k_0 R \sim \sqrt{2\varepsilon}$ produces the first-order reduction of axial penetration toward the sleeve. The auxiliary solution (4.1) and the illustrative evolution (5.3) approximate the nonlinear moving-front problem; higher-mode coupling and the actual curved-front boundary condition change the curvature already at leading order in ε and are treated in Section 7. The calibration (6.2) is appropriate when a separate model or measurement supplies a more reliable *on-axis* trajectory and one wants to retain the Bessel-predicted bending.

9 Selected Bessel identities

$$J_0'(x) = -J_1(x), \qquad \frac{d}{dx}[x^\nu J_\nu(x)] = x^\nu J_{\nu-1}(x), \tag{9.1}$$

$$J_\nu'(x) = \tfrac{1}{2}[J_{\nu-1}(x) - J_{\nu+1}(x)], \qquad J_0(x) = 1 - \frac{x^2}{4} + O(x^4). \tag{9.2}$$

The norm (3.8) is the appropriate Robin-boundary orthogonality formula for this problem.

10 Wall albedo: definition versus inference

The wall albedo is the fraction of incident radiative flux re-emitted or reflected back into the foam, in the effective boundary model:

$$a = \frac{F_{\text{returned}}}{F_{\text{incident}}}, \qquad 0 \le a \le 1, \qquad F_{\text{net loss}} = (1-a)F_{\text{incident}}. \tag{10.1}$$

Consequently the characteristic Robin parameter determines only the combination

$$1 - a = \frac{4\varepsilon}{3\rho\kappa R}. \tag{10.2}$$

Measured outgoing temperatures and an independently estimated wall heating rate are not, by themselves, sufficient to justify an algebraic ratio such as $(F_0 + \dot{E}_w/2)/(F_0 - \dot{E}_w/2)$ for albedo. A detector/transport energy balance defining which flux is incident and returned is required before inferring $a(t)$. If a is time dependent, the eigenbasis is also time dependent and the constant-ε modal derivation needs revision.

References

[1] O. A. Hurricane and J. H. Hammer, "Bent Marshak waves," *Physics of Plasmas* **13**, 113303 (2006), doi:10.1063/1.2388268.

[2] J. H. Hammer and M. D. Rosen, *Physics of Plasmas* **10**, 1829 (2003).